Chapter 26: Amniote Origins and Nonavian Reptiles

Origin and Early Evolution of Amniotes

- **Amniotes** are a monophyletic group that arose from **anthracosaurs** (anamniotes) in the early Carboniferous period.
- **Diadectes** is the likely sister taxon of amniotes.
- Early amniotes were small and lizardlike, radiating by the early Permian.

Figure 1: Evolution of amniotes. The evolutionary origin of amniotes occurred by evolution of an amniotic egg that made reproduction on land possible, although this egg may well have developed before the earliest amniotes had ventured far on land. The amniote assemblage, which includes nonavian reptiles, birds, and mammals, evolved from a lineage of small, lizardlike forms that retained the anapsid skull pattern of early tetrapods. First to diverge from the primitive stock was a lineage that evolved a skull pattern termed the synapsid condition. All other amniotes, including birds and all living nonavian reptiles except turtles, have a skull pattern known as diapsid. Turtles retain the primitive, anapsid skull pattern. The great Mesozoic radiation of reptiles may have resulted partly from the increased variety of ecological habitats that amniotes could exploit.

- **Skull Fenestration Patterns**:
  - **Anapsid**: No temporal openings behind the orbit (ancestral condition, turtles).
  - **Diapsid**: Two temporal openings (one pair low on cheeks, one pair dorsal). Found in birds and all traditional “reptiles” except turtles.
    - **Lepidosaurs**: Lizards, snakes, tuataras.
    - **Archosaurs**: Dinosaurs, pterosaurs, crocodilians, birds.
    - **Sauropterygians**: Extinct aquatic groups (e.g., plesiosaurs).
    - **Ichthyosaurs**: Extinct dolphinlike forms.
    - **Turtles**: Classification controversial; may be diapsids that lost openings.

- **Synapsid**: Single pair of temporal openings low on cheeks (mammals and their ancestors).

Figure 2: Cladogram of living Amniota showing monophyletic groups. Some shared derived characters (synapomorphies) of the groups are given. The skulls represent the ancestral condition of the three groups. Skulls of modern diapsids and synapsids are often highly modified by loss or fusion of skull bones that obscures the ancestral condition. Representative skulls for anapsids are **Nyctiphruetus** of the upper Permian; for diapsids, **Youngina** of the upper Permian; for synapsids, **Aerosaurus**, a pelycosaur of the lower Permian. The relationship of turtles to other reptiles is controversial: some researchers consider them archosaurs; others consider them lepidosaurs or direct living descendents of anapsids.

- **Derived Characters of Amniotes**:
  - **Amniotic Egg**: Four extraembryonic membranes.
    - **Amnion**: Encloses embryo in fluid (cushioning, aqueous medium).
    - **Allantois**: Stores metabolic wastes; respiratory surface.
    - **Chorion**: Surrounds contents; highly vascularized for respiration.
    - **Yolk sac**: Nutrient storage.
    - Shell: Mineralized/flexible; mechanical support; semipermeable barrier.
  - **Rib Ventilation**: Aspiration (drawing air in) by expanding thoracic cavity using costal muscles (negative pressure breathing).
  - **Thicker, Waterproof Skin**: Keratinized (beta keratin in reptiles), lipid-rich to limit water loss; not used for respiration.

Figure 3: Amniotic egg. The embryo develops within the amnion and is cushioned by amniotic fluid. Food is provided by yolk from the yolk sac and metabolic wastes are deposited within the allantois. As development proceeds, the allantois fuses with the chorion, a membrane lying against the inner surface of the shell; both membranes are supplied with blood vessels that assist in the exchange of oxygen and carbon dioxide across the porous shell. Because this kind of egg is an enclosed, self-contained system, it is often called a “cleidoic” egg (Gr. **kleidoun**, to lock in).

Characteristics of Nonavian Reptiles

- **Distinctions from Amphibians**:
  - **Skin**: Tough, dry, scaly skin protects against desiccation and physical injury.
  - Scales are keratinized (beta keratin) and derived from epidermis (unlike fish scales).
  - Skin is not used for respiration (unlike amphibians).

Figure 4: Section of the skin of a nonavian reptile showing overlapping, keratinized scales in the epidermis and bony osteoderms in the dermis.

- **Physiological Adaptations**:
  - **Circulatory System**: Higher blood pressure; functionally separate pulmonary and systemic circuits. Incomplete ventricular septum (in non-crocodilians) allows shunting of blood during diving or apnea.
  - **Water Conservation**: Excretion of nitrogenous waste as **uric acid** (semisolid paste) to conserve water.
  - **Nervous System**: Enlarged cerebrum; advanced connections to sensory systems.

Figure 5: Internal structure of a male crocodile.

Characteristics and Natural History of Reptilian Orders

Order Testudines (Chelonia): Turtles

- **Shell Structure**: Unique among vertebrates; dorsal **carapace** and ventral **plastron**.
- The shell consists of an outer layer of keratin and an inner layer of bone, formed by the fusion of ribs and vertebrae.
- **Limbs**: Uniquely located inside the ribs.

Figure 6: Skeleton and shell of a turtle, showing fusion of vertebrae and ribs with the carapace. The long and flexible neck allows the turtle to withdraw its head into its shell for protection.

- **Feeding**: Turtles lack teeth; their jaws are covered by tough, keratinized plates forming a beak.
- **Respiration**: Rigid shell prevents chest expansion; they use abdominal and pectoral muscles as a “diaphragm” to breathe. Aquatic forms may pump water in mouth/cloaca.

Figure 7: Snapping turtle, **Chelydra serpentina**, showing the absence of teeth. Instead, the jaw edges are covered with a keratinized plate.

- **Reproduction**: All turtles are oviparous with internal fertilization.
- **Mating Behavior**: Males may be smaller; concave plastron in some terrestrial species facilitates mounting.
- **Senses**: Poor hearing (mute); good sense of smell and color vision.

Figure 8: Mating Galápagos tortoises, **Geochelone elephantopus**. The male has a concave plastron that fits over the highly convex carapace of the female, helping to provide stability during mating. Males utter a roaring sound during mating, the only time they are known to emit vocalizations.

- **Temperature-Dependent Sex Determination (TSD)**: In many turtles (and some other reptiles), nest temperature determines sex.
- Typically, low temperatures produce males, and high temperatures produce females.

Figure 9: Temperature-dependent sex determination in the European pond turtle, **Emys orbicularis**. Eggs incubated at high temperatures produce females, while eggs incubated at low temperature produce males.

- **Diversity and Ecology**:
  - **Sea Turtles**: Large, marine forms with paddle-like limbs; return to land only to lay eggs.

Figure 10: Green sea turtle, **Chelonia mydas**. Green turtles are herbivores that subsist on marine grasses and algae. Sea turtles range widely in the oceans, returning to land only to deposit their eggs. Sea turtles are found in all tropical oceans.

- **Predation Strategies**: Some turtles, like the Alligator Snapping Turtle, use lures (e.g., tongue appendage) to attract prey.

Figure 11: Alligator snapping turtle, **Macroclemys temmincki**, of the southeastern United States lies on the bottom, mouth agape, luring fishes and other unwary prey by undulating a pink, wormlike protrusion from its tongue. Any prey attempting to eat the bait is instantly captured in powerful jaws.

Order Squamata: Lizards and Snakes

- **Characteristics**:
  - 95% of living nonavian reptiles.
  - **Kinetic skull**: Movable joints (quadrate) allow snout tilting and seizing/manipulating prey.
  - **Viviparity**: Evolved 100 times; associated with cool climates (retention of young for thermoregulation).

Figure 12: Kinetic diapsid skull of a modern lizard (monitor lizard, **Varanus** sp.) showing the joints that allow the snout and upper jaw to move on the rest of the skull. The quadrate can move at its dorsal end and ventrally at both the lower jaw and the pterygoid. The front part of the braincase is also flexible, allowing the snout to be raised. Note that the lower temporal opening is very large with no lower border; this modification of the diapsid condition, common in modern lizards, provides space for expansion of large jaw muscles. The upper temporal opening lies dorsal and medial to the postorbital-squamosal arch and is not visible in this drawing.

Suborder Sauria: Lizards

- **Diversity**: Lizards are an extremely diverse group, including geckos, iguanids, skinks, monitors, and chameleons.
- **Senses**: Most have movable eyelids and an external ear opening (unlike snakes). Vision is usually keen (daylight).
- **Water Conservation**: Excrete semisolid urine (uric acid) to conserve water. Some store fat in tails.
- **Ectothermy**: Allows survival in low-productivity ecosystems (deserts).
- **Geckos**: Small, agile, nocturnal lizards with adhesive toe pads; known for vocalizations.

Figure 13: Tokay, **Gekko gecko**, of Southeast Asia has a true voice and is named after the strident repeated **to-kay, to-kay** call.

- **Iguanids**: Diverse group; includes the Marine Iguana, which feeds on algae underwater and has salt glands.

Figure 14: A large male marine iguana, **Amblyrhynchus cristatus**, of the Galápagos Islands, feeding underwater on algae. This is the only marine lizard in the world. It has special salt-removing glands in the eye orbits and long claws that enable it to cling to the bottom while feeding on small red and green algae, its principal diet. It may dive to depths exceeding 10 m (33 feet) and remain submerged more than 30 minutes.

- **Chameleons**: Arboreal specialists with zygodactylous feet, independent eye movement, and projectile tongues for catching prey.

Figure 15: A chameleon snares a dragonfly. After cautiously edging close to its target, the chameleon suddenly lunges forward, anchoring its tail and feet to the branch. A split second later, it launches its sticky-tipped, foot-long tongue to trap the prey. The eyes of this common European chameleon, **Chamaeleo chamaeleon**, are swiveled forward to provide binocular vision and excellent depth perception.

- **Legless Lizards**: Some lizards, like glass lizards, have lost limbs but retain movable eyelids and external ears (distinguishing them from snakes).

Figure 16: A glass lizard, **Ophisaurus** sp., of the southeastern United States. This legless lizard feels stiff and brittle to the touch and has an extremely long, fragile tail that readily fractures when the animal is struck or seized. Most specimens, such as this one, have only a partly regenerated tip to replace a much longer tail previously lost. Glass lizards can be readily distinguished from snakes by the deep, flexible groove running along each side of the body. They feed on worms, insects, spiders, birds' eggs, and small nonavian reptiles.

- **Venomous Lizards**: The Gila monster and Mexican beaded lizard are the only venomous lizards; venom is chewed into the wound from glands in the lower jaw.

Figure 17: Gila monster, **Heloderma suspectum**, of southwestern United States desert regions and the related Mexican beaded lizard are the only venomous lizards known. These brightly colored, clumsy-looking lizards feed principally on birds' eggs, nesting birds, mammals, and insects. Unlike venomous snakes, the Gila monster secretes venom from glands in its lower jaw. The chewing bite is painful to humans but seldom fatal.

- **Amphisbaenians** ("Worm Lizards"): Highly specialized for burrowing (fossorial).
- They have elongate, cylindrical bodies, reduced or absent limbs, and solid skulls for digging.

Figure 18: A worm lizard of the suborder Amphisbaenia. Worm lizards are burrowing forms with a solidly constructed skull used as a digging tool. The species pictured, **Amphisbaena alba**, is widely distributed in South America.

Suborder Serpentes: Snakes

- **Characteristics**: Limbless; lack pectoral/pelvic girdles (vestiges in pythons/boas); numerous vertebrae.
- **Feeding Adaptations**: Highly kinetic skull; mandibles joined only by muscle and skin, allowing wide expansion to swallow large prey.

Figure 19: A, Lateral view of python skull. Each side of the extremely kinetic skull has several movable bones (labeled) which permit extraordinary movements of the jaw in feeding. The halves of the lower jaw are united by flexible soft tissues, permitting wide separation and independent movement of each side. B, The great mobility of the snake jaw and skull elements is evident in this snake swallowing an egg.

- **Arboreal Adaptations**: Some snakes, like the parrot snake, have slender bodies for moving through trees.

Figure 20: Parrot snake, **Leptophis ahaetulla**. The slender body of this Central American tree snake is an adaptation for sliding along branches without weighing them down.

- **Sensory Systems**:
  - **Vision**: Eyelids are fused into a transparent **spectacle**.
  - **Hearing**: No external ears; internal ears detect low frequencies and ground vibrations.
  - **Chemoreception**: **Jacobson's organs** (vomeronasal organs) in the roof of the mouth are used with the forked tongue to smell the environment.

Figure 21: A blacktail rattlesnake, **Crotalus molossus**, flicks its tongue to smell its surroundings. Scent particles trapped on the tongue's surface are transferred to Jacobson's organs, olfactory organs in the roof of the mouth. Note the heat-sensitive pit organ between the nostril and eye.

- **Locomotion**:
  - **Lateral undulation**: S-shaped path, pushing against irregularities (most common).

- **Concertina**: Used in narrow passages; bracing loops against walls.
- **Rectilinear**: Straight line movement using belly scales; used by heavy snakes.
- **Side-winding**: Used on loose sand; throwing body forward in loops.

Figure 22: Snake locomotion. A, Concertina motion. B, Lateral undulation. C, Rectilinear motion. D, Sidewinder motion. Refer to text for explanation.

- **Predation**:
  - **Constriction**: Suffocating prey by tightening coils (e.g., boas, pythons, some colubrids).

Figure 23: Nonvenomous African house snake, **Boaedon fuluginosus**, constricting a mouse before swallowing it.

- **Venom**: Delivered via fangs.
  - **Elapids**: Permanently erect fangs (cobras, mambas, coral snakes).

Figure 24: Spectacled, or Indian, cobra, **Naja naja**. Cobras erect the front of the body and flatten the neck as a threat display and before attacking. Although a cobra's strike range is limited, all cobras are dangerous because of the extreme toxicity of their venom.

- **Vipers**: Movable tubular fangs that fold back when mouth is closed.
- **Pit Vipers**: Possess heat-sensitive **pit organs** between nostrils and eyes for detecting warm-blooded prey.

Figure 25: Pit organ of rattlesnake, a pit viper. Cutaway shows location of a deep membrane that divides the pit into inner and outer chambers. Heat-sensitive nerve endings are concentrated in the membrane.

- **Venom Types**: **Neurotoxic** (affects nervous system, respiratory paralysis) and **Hemorrhagin** (destroys blood and tissue).

Figure 26: Head of rattlesnake showing the venom apparatus. The venom gland, a modified salivary gland, is connected by a duct to the hollow fang.

Order Sphenodonta: Tuataras

- **Sphenodon**: Sole survivors of sphenodontid lineage (New Zealand).
- **Characteristics**: Diapsid skull with two complete arches; well-developed median **parietal eye** (light sensitive); slow growth/low reproductive rates.

Figure 27: Tuatara, **Sphenodon** sp., a living representative of order Sphenodonta. This "living fossil" reptile has, on top of the head, a well-developed parietal "eye" with retina, lens, and nervous connections to the brain. Although covered with scales, this third eye is sensitive to light. The parietal eye may have been an important sense organ in early reptiles. Tuataras are found today only on certain islands off the coastline of New Zealand.

Order Crocodilia: Crocodiles and Alligators

- **Lineage**: Archosaurs (surviving lineage along with birds).

- **Characteristics**: Elongate, robust skull; **thecodont** dentition (teeth in sockets); complete **secondary palate** (breathe while mouth full); four-chambered heart.
- **Behavior**: Parental care (guarding nest, opening nest for hatchlings); vocalizations.
- **Sex Determination**: Temperature-dependent (low = females, high = males).
- **Distinction**: Crocodiles (narrow snout, 4th lower tooth visible); Alligators (broader snout, 4th tooth hidden).

Figure 28: Crocodilians. A, Nile crocodile, **Crocodylus niloticus**, basking. The fourth tooth of the lower jaw fits outside the slender upper jaw; alligators lack this feature. B, American alligator, **Alligator mississippiensis**, an increasingly noticeable resident of rivers, bayous, and swamps of the southeastern United States.